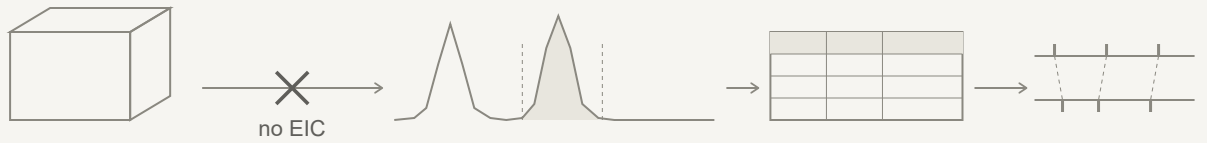
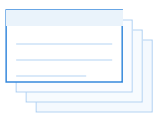
(a) Peak-centric route — not used



(b) Spectral route — this work

MSGeter (blue) · DMetFinder (green)



group A · 4 files

one file = one sample

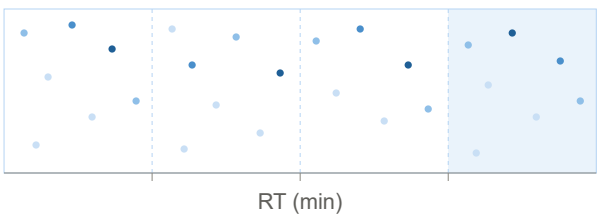
every sample handled identically



group B · 4 files

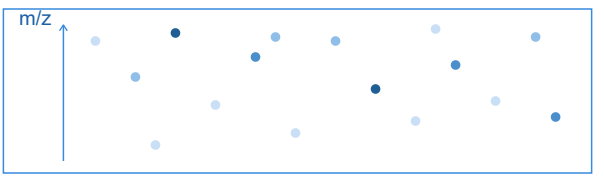
scans binned by minute

floor(RT) → one bin per minute



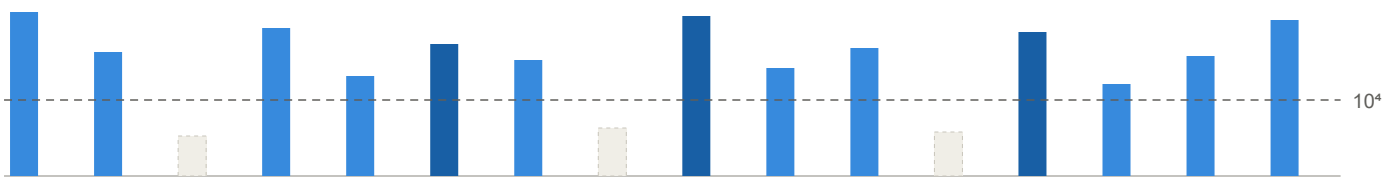
inside one bin

y = m/z · darker = stronger



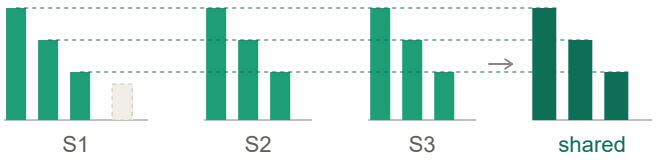
one bar per RT & m/z

height = intensity · dashed bars fall below the floor and are dropped



align within group

only ions that recur across files survive



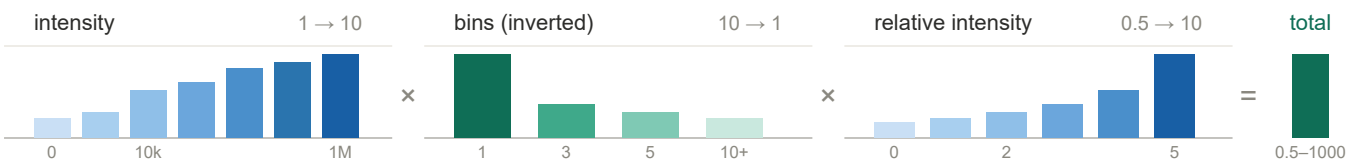
two thresholds, two jobs

- 50% of files → candidate ions
- 80% of files → ions compared

candidates are the set that gets scored

score every candidate ion

total = intensity × count × relative intensity · the same three tables also score every ion of the run

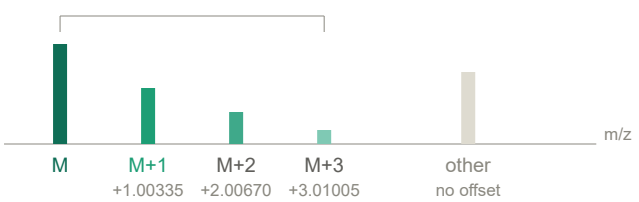


count = how many of the 18 minute bins carry this m/z in one sample; inverted, so 1–2 bins score 10 and ≥10 bins score 1

isotope check on every ion

M+1, M+2 and M+3 are searched at three fixed mass offsets in the same minute

one ion at m/z M



abundance gate, m/z 100–1000

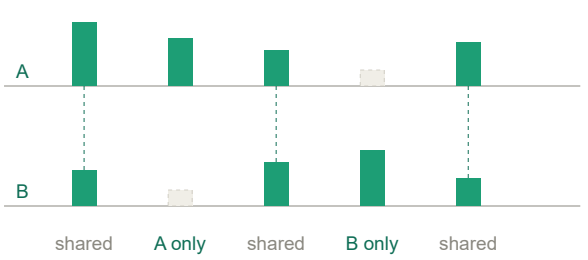
M / M+1	1 – 50	M+1 is an isotope
M+1 / M+2	> 1 and 1 – 100	M+2 is an isotope
M+2 / M+3	> 1 and 1 – 100	M+3 is an isotope

for m/z ≥ 1000 the window is 0.5 – 5

a gated ion is a heavy isotope, not a new feature;
the flag is carried into every row of the table below

identify differential ions between groups

both group vectors on one RT & m/z grid; the table is written in both directions



paired when the minute matches and |Δm/z| ≤ 0.003
dashed tile = no ion there; B-only rows go to the mirror sheet

differential ion table · one row per 80% ion of A

m/z	count A/B	avg A/B	M+1	call
267.1231	9 / 0	0	–	A only
214.0864	8 / 8	1.9	–	shared
219.0899	8 / 8	2.1	1	flagged isotope
301.1077	7 / 7	0.8	–	shared
336.1590	5 / 5	3.4	–	shared

count = files of the group carrying the ion
avg A/B = mean intensity ratio · 0 = partner missing
the call uses presence and abundance only; scores do not enter it

every ion keeps its m/z and its minute — no chromatographic peak is ever built